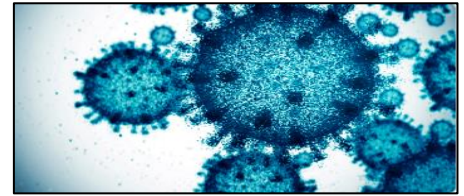


Covid-19 Machine Learning (ML) Project: Predicting new cases of Covid-19

Shlomi Jakubowicz.

jshlomi@yahoo.com or jshlomi81@gmail.com



<https://www.cdc.gov/covid/about/index.html>

Project Overview

In this project, we aim to develop a predictive model for the “new_cases_smoothed_per_million” metric using the comprehensive COVID-19 dataset provided by Our World in Data. This dataset includes a wide range of indicators related to the COVID-19 pandemic, such as case counts, testing rates, and vaccination data, which are crucial for understanding the spread and impact of the virus globally.

The target feature, "new_cases_smoothed_per_million", is defined as "New confirmed cases of COVID-19 (7-day smoothed) per 1,000,000 people." Counts may include probable cases, where reported. This feature is favorable because it reduces daily fluctuations and allows for accurate comparisons between regions with different population sizes.

Motivation

Predicting new COVID-19 cases is essential for several reasons:

- **Public Health Planning:** Accurate predictions help allocate resources like hospital beds, medical supplies, and healthcare personnel efficiently, ensuring healthcare systems are prepared for surges in cases.
- **Policy Making:** Governments can use these predictions to make informed decisions about public health measures such as lockdowns, travel restrictions, and social distancing guidelines, helping to control the spread of the virus.
- **Economic Stability:** Businesses and economic planners can anticipate disruptions and plan accordingly, minimizing economic losses and ensuring the continuity of essential services.
- **Public Awareness:** Providing the public with accurate forecasts enhances awareness and encourages compliance with health measures, leading to better community-level responses and reduced transmission.
- **Research and Development:** Accurate predictions guide researchers in understanding the virus's dynamics and evaluating the effectiveness of interventions, accelerating the development of new strategies to combat the pandemic.

By developing a robust predictive model for new COVID-19 cases, we can support these critical areas and contribute to managing and mitigating the impact of the pandemic.

The Our World In Data (OWID) Covid-19 Data Set

The dataset from Our World in Data spans from January 2020 to August 2024 and includes comprehensive metrics related to COVID-19. Composed of 429,435 records (rows) and 67 features (columns), the dataset provides:

- **Cases:** Total confirmed cases, new confirmed cases, and new cases per million people. These metrics are also available as seven-day rolling averages to smooth out daily fluctuations.
- **Deaths:** Total deaths attributed to COVID-19, new deaths, and new deaths per million people. These metrics are also available as seven-day rolling averages.
- **Testing:** Total tests conducted, new tests, and positivity rates. These metrics are also available per thousand people and as seven-day rolling averages.

- Hospitalizations: Number of COVID-19 patients in hospitals and ICUs, both daily and cumulative, including weekly admissions.
- Vaccinations: Total vaccination doses administered, number of people vaccinated, boosters. These metrics are also available per hundred people & as 7-day rolling averages.
- Population Metrics: Information on population size, density, median age, GDP per capita, and other demographic indicators (location, country iso code, continent).

The data set was downloaded from: <https://docs.owid.io/projects/covid/en/latest/dataset.html>

Table 1 OWID Covid-19 Features Table.

Category	Features
Cases	total_cases, new_cases, new_cases_smoothed, total_cases_per_million, new_cases_per_million, new_cases_smoothed_per_million
Deaths	total_deaths, new_deaths, new_deaths_smoothed, total_deaths_per_million, new_deaths_per_million, new_deaths_smoothed_per_million
Excess Mortality	excess_mortality, excess_mortality_cumulative, excess_mortality_cumulative_absolute, excess_mortality_cumulative_per_million
Hospitalizations	icu_patients, icu_patients_per_million, hosp_patients, hosp_patients_per_million, weekly_icu_admissions, weekly_icu_admissions_per_million, weekly_hosp_admissions, weekly_hosp_admissions_per_million
Testing	total_tests, new_tests, total_tests_per_thousand, new_tests_per_thousand, new_tests_smoothed, new_tests_smoothed_per_thousand, positive_rate, tests_per_case, tests_units
Vaccinations	total_vaccinations, people_vaccinated, people_fully_vaccinated, total_boosters, new_vaccinations, new_vaccinations_smoothed, total_vaccinations_per_hundred, people_vaccinated_per_hundred, people_fully_vaccinated_per_hundred, total_boosters_per_hundred, new_vaccinations_smoothed_per_million, new_people_vaccinated_smoothed, new_people_vaccinated_smoothed_per_hundred
Population Metrics, Policy Responses, Date Other	iso_code, continent, location, date, population, population_density, median_age, aged_65_older, aged_70_older, gdp_per_capita, extreme_poverty, cardiovasc_death_rate, diabetes_prevalence, female_smokers, male_smokers, handwashing_facilities, hospital_beds_per_thousand, life_expectancy, human_development_index, reproduction_rate, date, stringency_index

Data Preparation and EDA

Data transformation: converted the “date” Object feature to a datetime format.

In the data cleaning process, rows where the 'continent' column is NaN are removed. This step is taken to exclude overall metrics for various regions, income groups, and global totals, ensuring the dataset focuses on specific countries and their respective data.

Values count for “new_cases_smoothed_per_million” (Target):

The target feature contains approximately 30% records with a value of 0 cases and has a low percentage of missing values, around 3%.

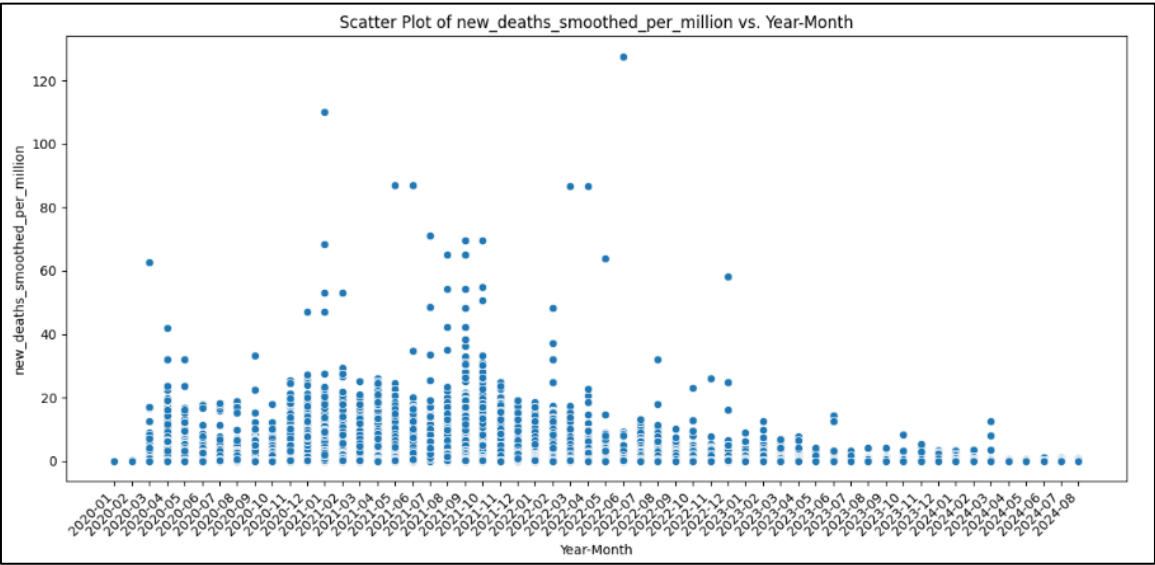


Figure 1 The target value (new_cases_smoothed_per_million) over time (Year-Month).

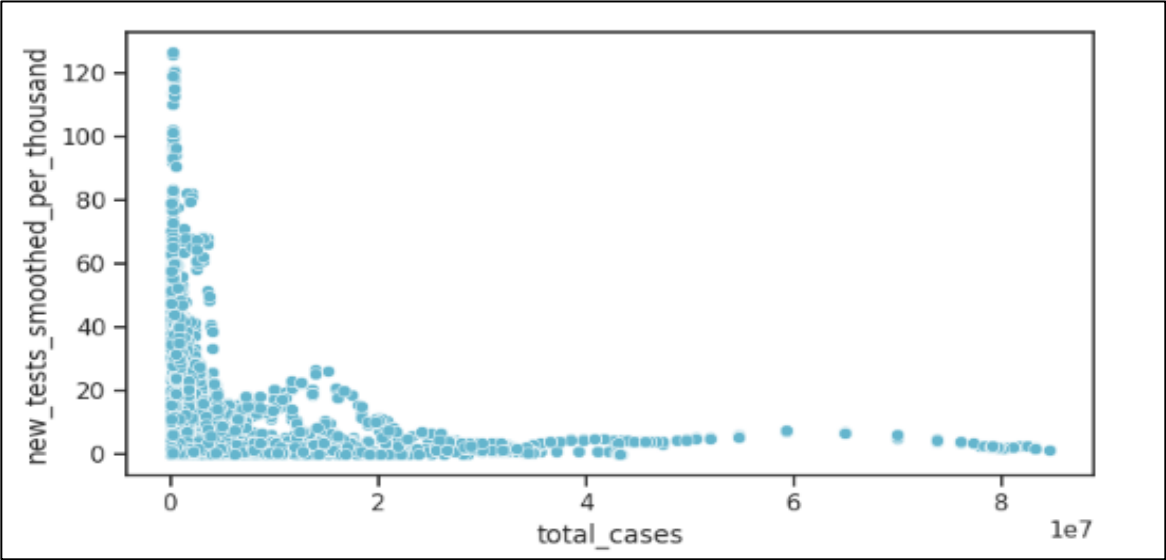


Figure 2 The target value (new_cases_smoothed_per_million) distribution.

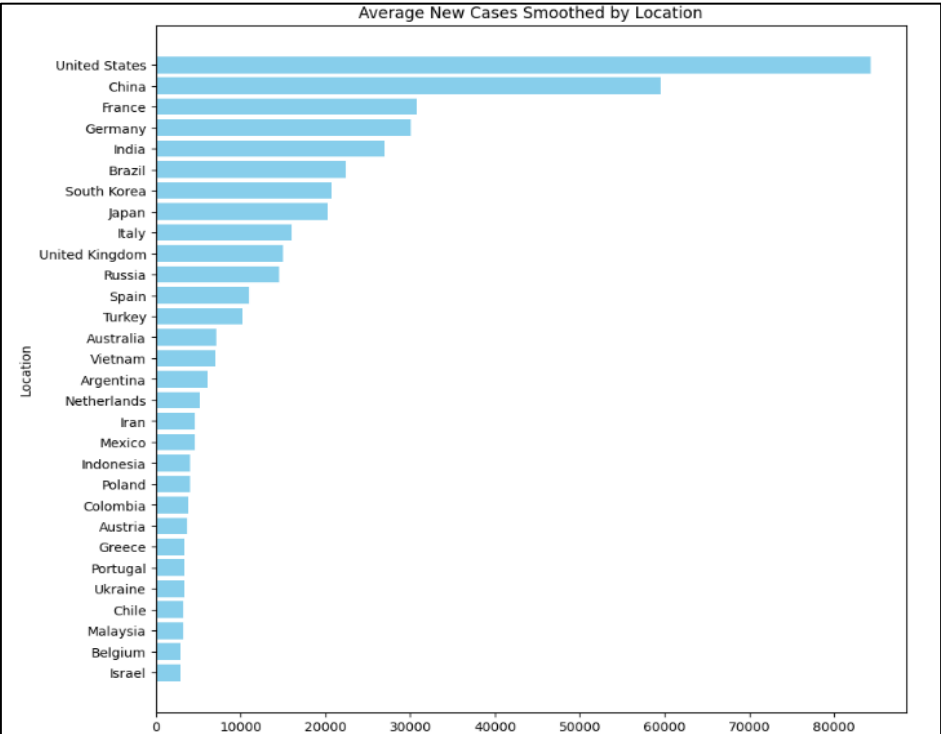


Figure 3 Average new_cases_smoothed by Location

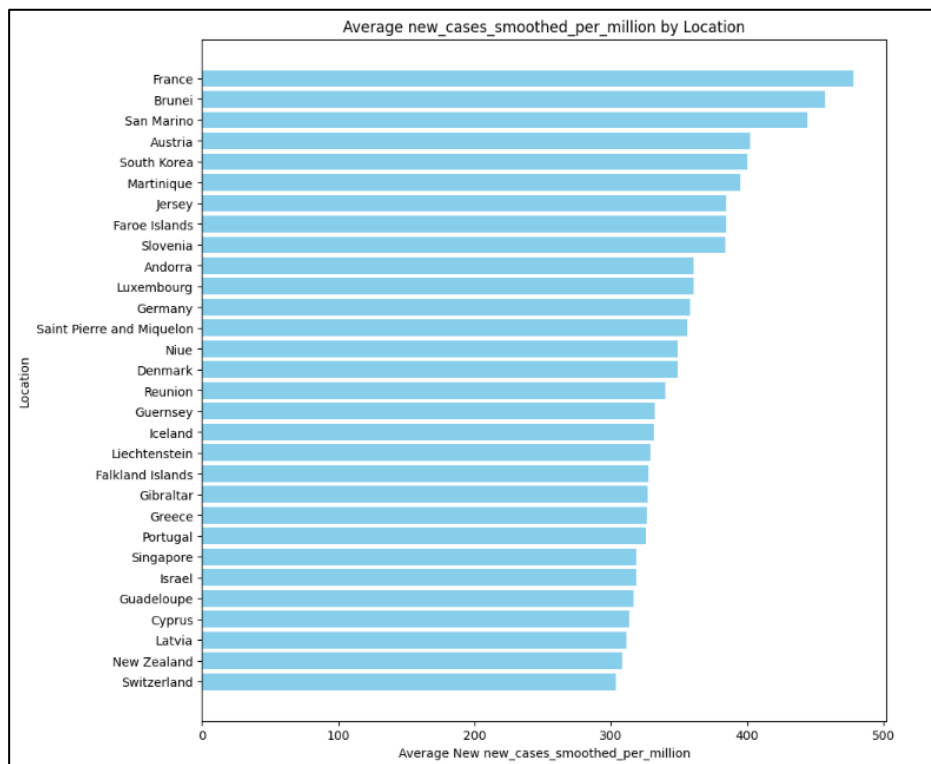


Figure 4 Average new_cases_smoothed_per_million by Location

Estimated Key Factors Impacting COVID-19 Spread (2020-2023)

An analysis of key features impacting new COVID-19 cases from 2020 to 2023 aimed to understand how socio-economic and health-related factors influenced the spread of COVID-19, measured as new cases smoothed per million. It was estimated these key features would impact the spread. By categorizing (low, medium, high) and statistically analyzing the stringency index, population density, GDP per capita, and vaccination rates, various statistical tests determined relationships and differences among these factors.

Findings:

- Stringency Index: Higher stringency leads to fewer cases, as stricter measures reduce transmission.
- Population Density: Higher density results in more cases, due to faster virus spread.
- GDP per Capita: Higher GDP correlates with more cases, likely due to increased mobility and better reporting.
- Vaccination Rate: Medium rates show the highest cases, possibly due to efficacy, distribution challenges, and reporting standards.

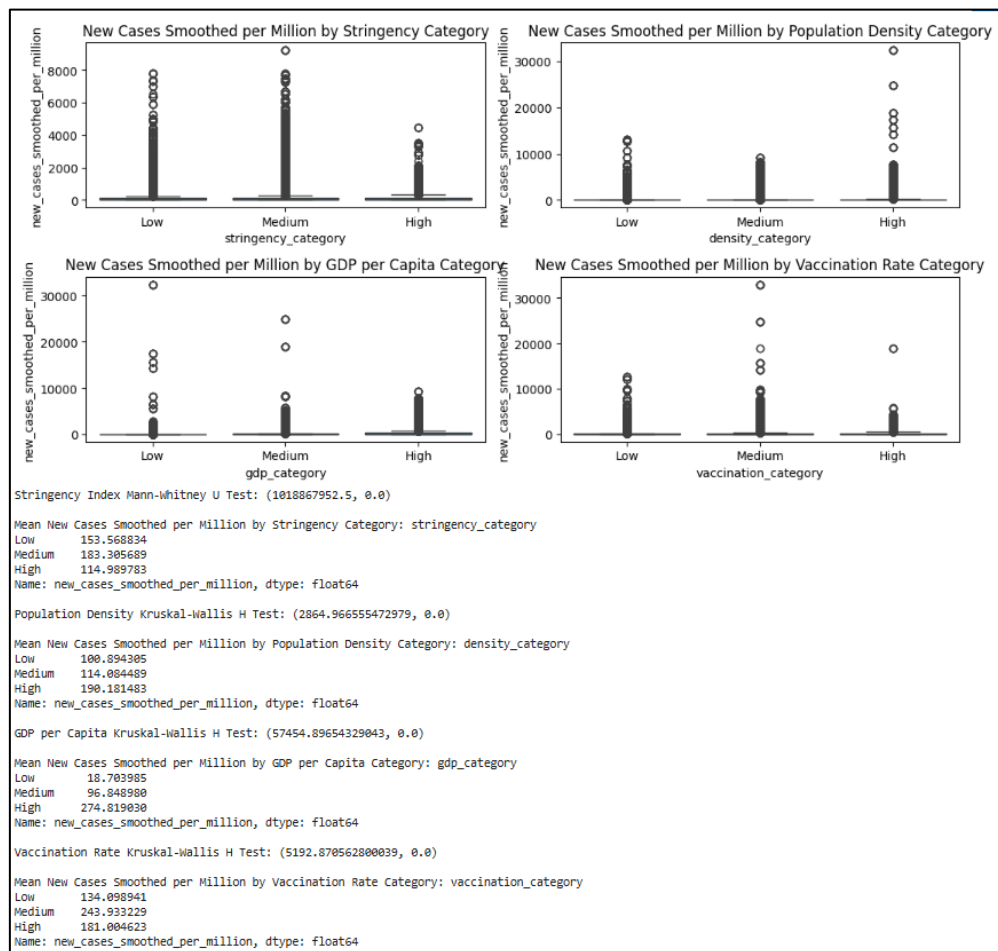


Figure 5 Estimated Key Factors Impacting COVID-19 Spread (2020-2023)

Data Skewness Test

Most of the features (and target) were determined to be skewed (was also seen using a box plot). Indicating the need for a non-parametric distribution and usage of IQR for outlier detection. Including a ~16% for the target value (New cases smooth per million).

	skewness
new_cases	340.480082
new_cases_smoothed	128.963316
new_cases_per_million	58.644440
new_deaths	46.585021
new_tests	43.835708
new_deaths_per_million	29.637527
new_cases_smoothed_per_million	22.916542

Figure 6 Features Skewness (Not all features are shown)

	Outlier count	Percent
new_deaths_smoothed	72264.0	17.935519
new_cases_smoothed	71495.0	17.744658
new_deaths_smoothed_per_million	68887.0	17.097367
new_cases_smoothed_per_million	63126.0	15.667519
total_deaths	61106.0	15.166166
total_cases	59253.0	14.706262
population	43225.0	10.728202
total_cases_per_million	40785.0	10.122608
new_cases_per_million	38929.0	9.661959
new_cases	38929.0	9.661959
population_density	34255.0	8.501899
new_people_vaccinated_smoothed	28328.0	7.030851
new_vaccinations_smoothed	27125.0	6.732273
new_deaths	24937.0	6.189223
new_deaths_per_million	24937.0	6.189223
new_people_vaccinated_smoothed_per_hundred	22063.0	5.475913
total_deaths_per_million	19903.0	4.939813
new_vaccinations_smoothed_per_million	15795.0	3.920230
extreme_poverty	15066.0	3.739297
gdp_per_capita	14187.0	3.521134
new_tests_smoothed	13684.0	3.396292
diabetes_prevalence	13392.0	3.323819
tests_per_case	12715.0	3.155792
reproduction_rate	12375.0	3.071406
total_tests	11551.0	2.866893
new_tests_smoothed_per_thousand	10471.0	2.598843
hospital_beds_per_thousand	10044.0	2.492864

Figure 7 Outliers Percentages (Not all features are shown)

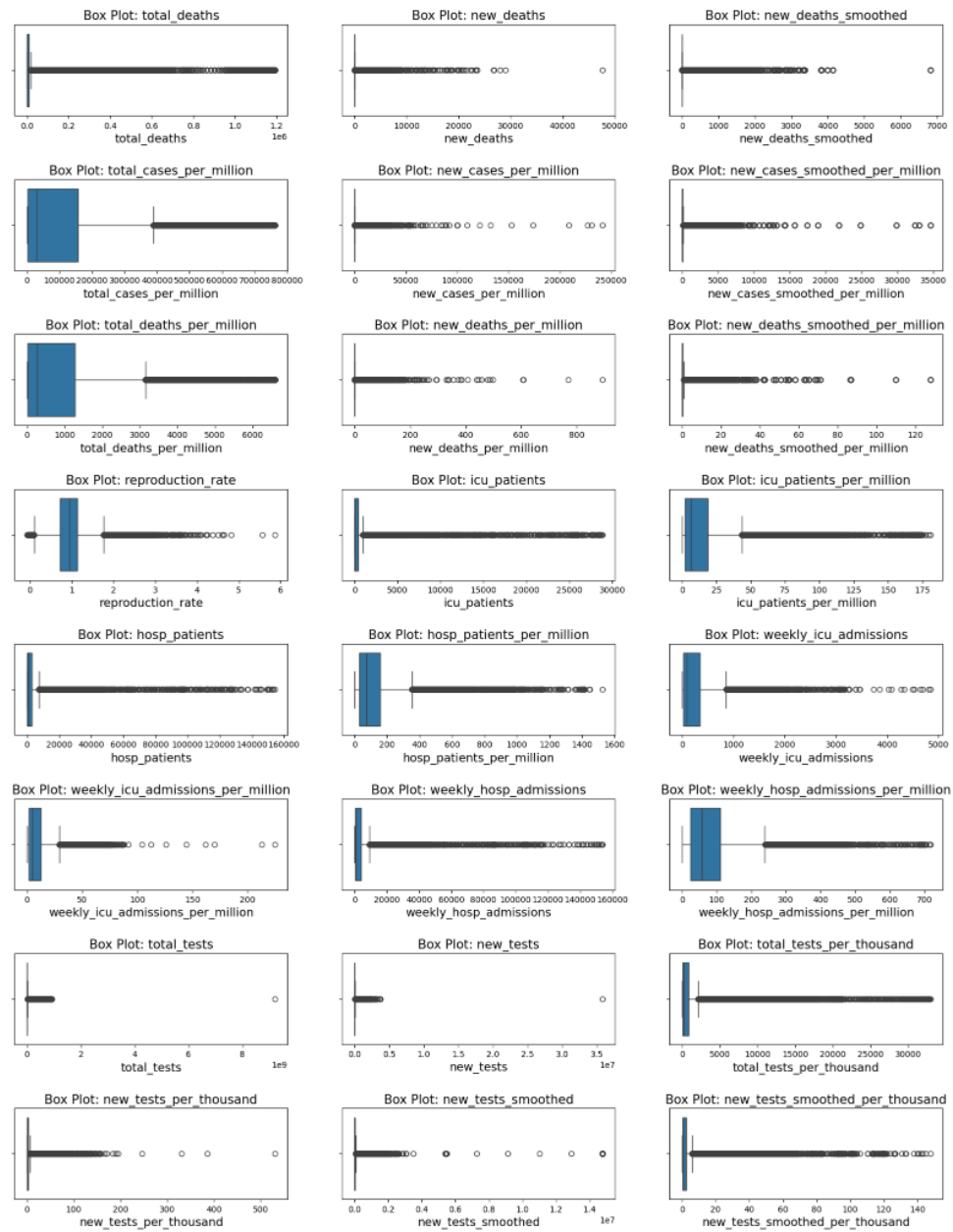
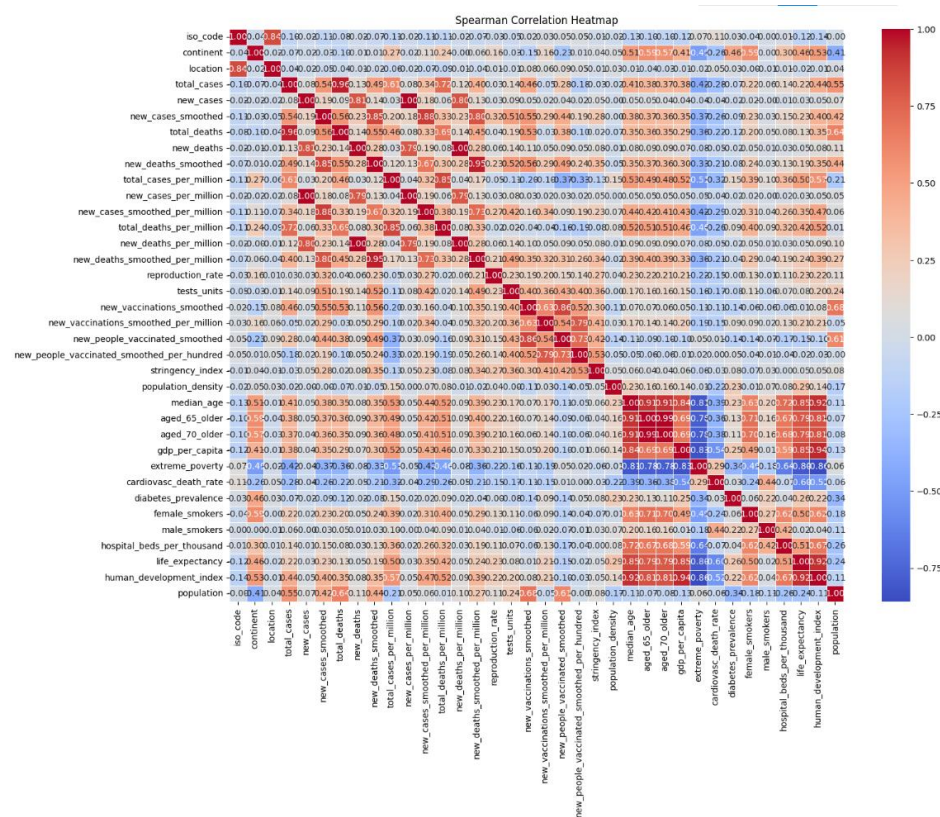


Figure 8 BOX-Plots (Not all features are shown)

Correlation

As most of the data is not normally distributed a Spearman correlation was performed to see potential contention between features.



Outliers Handling & Handling Missing Values

To handle outliers in features that are not normally distributed, I applied a log transformation to the numerical parameters and used Interquartile Range (IQR) detection. Detected outliers were set to NaN and then imputed along with other missing values using the MICE imputer.

	Outlier count	Percent
new_deaths_smoothed_per_million	61299.0	15.214068
new_deaths_smoothed	44269.0	10.987317
new_cases_per_million	38929.0	9.661959
new_cases	38929.0	9.661959
total_cases	33708.0	8.366136
total_cases_per_million	31964.0	7.933285
new_deaths	24937.0	6.189223
new_deaths_per_million	24937.0	6.189223
new_people_vaccinated_smoothed_per_hundred	21049.0	5.224244
population_density	9145.0	2.269738
new_vaccinations_smoothed_per_million	5418.0	1.344717
male_smokers	3348.0	0.830955
population	3348.0	0.830955
diabetes_prevalence	3348.0	0.830955
new_cases_smoothed	49.0	0.012162
new_cases_smoothed_per_million	35.0	0.006687
new_vaccinations_smoothed	16.0	0.003971
total_deaths	0.0	0.000000
total_deaths_per_million	0.0	0.000000
new_people_vaccinated_smoothed	0.0	0.000000
gdp_per_capita	0.0	0.000000
cardiovasc_death_rate	0.0	0.000000
extreme_poverty	0.0	0.000000
female_smokers	0.0	0.000000
hospital_beds_per_thousand	0.0	0.000000

Figure 9 Outliers Percentages after Log Transformation.

Feature Engineering, Feature Selection, and Data Leakage Prevention

The feature engineering process begins with the `prepare_covid_data` function, transforming raw COVID-19 data for time-series forecasting. Key steps include creating the `total_cases_change_7` feature, representing the change in total cases over the last 7 days, excluding the current day. This feature assumes that the number of cases in the past week is a strong indicator for future case projections. Time-based attributes such as year, month, week, day, and day of the week are extracted from the date column. The target variable is `new_cases_smoothed_per_million`.

To prevent data leakage, columns like `total_cases` and `total_cases_per_million` are removed because they contain information not available at prediction time. All new cases features, except the target variable, are dropped to ensure only relevant information is used.

Four (4) feature selection techniques are applied:

1. Lasso (L1 Regularization) shrinks coefficients of less important features towards zero.
2. Ridge (L2 Regularization) reduces the magnitude of coefficients.
3. Gradient Boosting Regressor provides feature importances based on prediction error reduction.
4. Random Forest Regressor averages feature contributions across decision trees.

Each model is trained on scaled feature data, identifying important features based on coefficient magnitude or feature importance scores.

The results from all four methods are compiled into a Pandas DataFrame (`selection_df`), showing whether each feature was selected by each technique, with a 'Sum' column indicating the total number of methods that selected each feature.

Table 2 Feature Selection Output Table

Feature	Lasso	GradientBoost	RandomForest	Ridge	Sum
iso_code	1	1	1	1	4
continent	1	1	1	1	4
total_deaths	1	1	1	1	4
new_deaths	0	0	0	0	0
new_deaths_smoothed	1	1	1	1	4
total_deaths_per_million	1	1	1	1	4
new_deaths_per_million	0	0	0	0	0
new_deaths_smoothed_per_million	1	1	1	1	4
reproduction_rate	1	1	1	1	4
tests_units	1	1	1	1	4
new_vaccinations_smoothed	0	1	1	1	3
new_vaccinations_smoothed_per_million	1	1	1	1	4
new_people_vaccinated_smoothed	1	1	1	1	4
new_people_vaccinated_smoothed_per_hundred	1	1	1	1	4
stringency_index	1	1	1	1	4
population_density	1	1	1	1	4
median_age	1	0	1	1	3
aged_65_older	0	1	1	1	3
aged_70_older	1	1	1	1	4
gdp_per_capita	1	1	1	1	4
extreme_poverty	1	1	1	1	4
cardiovasc_death_rate	1	1	1	1	4
diabetes_prevalence	1	1	1	1	4
female_smokers	0	1	1	1	3
male_smokers	1	1	1	1	4
hospital_beds_per_thousand	1	1	1	1	4
life_expectancy	0	1	1	1	3
human_development_index	0	1	1	1	3
population	1	1	1	1	4
total_cases_change_7	1	1	1	1	4
year	1	1	1	1	4
month	1	1	1	1	4
week	1	1	1	1	4
day	0	0	1	1	2
dayofweek	0	1	1	1	3

5-Fold Cross-Validation and Regression Model Comparison for COVID-19 Forecasting

This code evaluates various regression models in predicting smoothed new COVID-19 cases per million. The workflow includes data preparation, data splitting, feature scaling, model training and evaluation, and comprehensive results reporting.

Data Preparation: The feature engineering process begins with the `prepare_covid_data` function, which engineers time-aware features and cleans the raw data, mitigating potential data leakage. A pre-selected feature set is loaded, filtered by time period, and categorical location identifiers are encoded.

Data Splitting: Data is split into folds based on unique geographical locations (`iso_code`). Models are trained on data from some locations and evaluated on unseen locations, ensuring realistic geographical generalization. Importantly, the `iso_code` is not used as a feature in the training models to avoid data leakage.

Feature Scaling: Numerical features are standardized within each fold using `StandardScaler`. Scaling parameters are learned from the training data to prevent information leakage and applied to validation and test sets.

Model Training and Evaluation: Various regression models (Dummy Regressor, Linear Regression, Random Forest Regressor, Gradient Boosting Regressor, SVR, K-Nearest Neighbors Regressor, XGBoost Regressor) are trained on scaled training data. Performance is evaluated using metrics: Mean Squared Error (MSE), Mean Absolute Error (MAE), R-squared (R2), Root Mean Squared Error (RMSE), and Root Mean Squared Logarithmic Error (RMSLE).

Results Reporting: Performance metrics for each model across all dataset splits within each fold are recorded and outputted to the console and an external text file. This detailed reporting facilitates a thorough comparison of each model's strengths and weaknesses in COVID-19 forecasting.

By using geographically data splitting and diverse evaluation metrics, this approach provides a robust comparison of regression models for predicting COVID-19 trends, emphasizing generalization to new geographical regions. Excluding direct location identifiers like `iso_code` from the training models encourages them to identify broadly applicable predictive relationships.

Table 3 Model Performance Summary (5-Fold CV Averages)

Model	Dataset	Avg. MSE	Avg. MAE	Avg. R2	Avg. RMSE	Avg. RMSLE	Rank
Random Forest Regressor	Train	0.0093	0.0259	0.9987	0.0951	0.0354	1
	Val	0.8560	0.5898	0.8604	0.9190	0.2520	
	Test	0.7602	0.5901	0.8743	0.8493	0.2786	
Gradient Boosting Regressor	Train	0.2409	0.3179	0.9653	0.4905	0.1622	2
	Val	0.6974	0.5567	0.8863	0.8318	0.2304	
	Test	0.6191	0.5366	0.9124	0.7848	0.2510	
XGBoost Regressor	Train	0.0316	0.1018	0.9954	0.1784	0.0557	3
	Val	0.6153	0.5767	0.8808	0.7726	0.2177	
	Test	0.6919	0.5671	0.9010	0.8303	0.2625	
SVR	Train	0.2246	0.2842	0.9679	0.4735	0.1585	4
	Val	1.0965	0.8005	0.8269	1.0476	0.2922	
	Test	1.3241	0.8928	0.8156	1.1598	0.3607	
Linear Regression	Train	1.7389	0.9898	0.7490	1.3186	0.3881	5
	Val	1.9767	1.0446	0.6890	1.4086	0.3996	
	Test	2.1181	1.0724	0.7098	1.4597	0.4230	
K-NNN Regressor	Train	0.0262	0.0476	0.9962	0.1620	0.0507	6
	Val	2.5379	1.1477	0.5845	1.5478	0.4085	
	Test	2.3981	1.1265	0.6297	1.5217	0.4234	
Dummy Regressor (Mean)	Train	6.9184	2.3016	0.0000	2.6309	0.7475	7
	Val	6.8459	2.2319	-0.0768	2.6187	0.7049	
	Test	7.1404	2.3097	-0.0032	2.6704	0.7836	

Conclusion

As can be seen in the 5-fold cross validation averages results table, the Random Forest Regressor consistently outperforms other models across most metrics, achieving the lowest average MSE, MAE, RMSE, and RMSLE on the training dataset. It also demonstrates strong performance on the validation and test datasets, making it the top-ranked model overall.

The Gradient Boosting Regressor and XGBoost Regressor also show competitive performance, particularly on the validation and test datasets, with Gradient Boosting Regressor achieving the highest R2 score on the test dataset. These models are ranked second and third, respectively.

SVR and Linear Regression models perform moderately well but are outperformed by ensemble methods like Random Forest and Gradient Boosting. K-Nearest Neighbors Regressor and Dummy Regressor (Mean) show relatively poor performance, with Dummy Regressor serving as a baseline model.

In conclusion, ensemble methods such as Random Forest and Gradient Boosting are highly effective for predicting smoothed new COVID-19 cases per million, with Random Forest Regressor being the most robust model overall.